

Chapter Review

Problems and Applications

1. Explain how each of the following developments would affect the supply of money, the demand for money, and the interest rate. Use diagrams to illustrate your answers.
 - a. The Fed's bond traders buy bonds in open-market operations.
 - b. An increase in credit-card availability reduces the amount of cash people want to hold.
 - c. The Fed reduces reserve requirements.
 - d. Households decide to hold more money to use for holiday shopping.
 - e. A wave of optimism boosts business investment and expands aggregate demand.
2. The Fed expands the money supply by 5 percent.
 - a. Use the theory of liquidity preference to illustrate in a graph the impact of this policy on the interest rate.
 - b. Use the model of aggregate demand and aggregate supply to illustrate the impact of this change in the interest rate on output and the price level in the short run.
 - c. When the economy makes the transition from its short-run equilibrium to its new long-run equilibrium, what happens to the price level?
 - d. How does this change in the price level affect the demand for money and the equilibrium interest rate?
 - e. Is this analysis consistent with the proposition that money has real effects in the short run but is neutral in the long run?

3. Suppose a computer virus disables the nation's automatic teller machines, making withdrawals from bank accounts less convenient. As a result, people want to keep more cash on hand, increasing the demand for money.
 - a. Assume the Fed does not change the money supply. According to the theory of liquidity preference, what happens to the interest rate? What happens to aggregate demand?
 - b. If instead the Fed wants to stabilize aggregate demand, how should it change the money supply?
 - c. If it wants to accomplish this change in the money supply using open-market operations, what should it do?
4. Consider two policies—a tax cut that lasts for only one year and a tax cut that is expected to be permanent. Which policy will stimulate greater spending by consumers? Which policy will have the greater impact on aggregate demand? Explain.
5. The economy is in a recession with high unemployment and low output.
 - a. Draw a graph of aggregate demand and aggregate supply to illustrate the current situation. Be sure to include the aggregate-demand curve, the short-run aggregate-supply curve, and the long-run aggregate-supply curve.
 - b. Identify an open-market operation that would restore the economy to its natural rate.
 - c. Draw a graph of the money market to illustrate the effect of this open-market operation. Show the resulting change in the interest rate.
 - d. Draw a graph similar to the one in [part a](#) to show the effect of the open-market operation on output and the price level. Explain in words why the policy has the effect that you have shown in the graph.
6. In the early 1980s, new legislation allowed banks to pay interest on checking deposits, which they could not do previously.
 - a. If we define money to include checking deposits, what effect did this legislation have on money demand? Explain.
 - b. If the Fed had maintained a constant money supply in the face of this change, what would have happened to the interest rate? What would have happened to aggregate demand and aggregate output?

- c. If the Fed had maintained a constant market interest rate (the interest rate on nonmonetary assets) in the face of this change, what change in the money supply would have been necessary? What would have happened to aggregate demand and aggregate output?
7. Suppose economists observe that an increase in government spending of \$10 billion raises the total demand for goods and services by \$30 billion.
- a. If these economists ignore the possibility of crowding out, what would they estimate the marginal propensity to consume (MPC) to be?
- b. Now suppose the economists allow for crowding out. Would their new estimate of the MPC be larger or smaller than their initial one?
8. An economy is producing output \$400 billion less than the natural level of output, and fiscal policymakers want to close this recessionary gap. The central bank agrees to adjust the money supply to hold the interest rate constant, so there is no crowding out. The marginal propensity to consume is $\frac{4}{5}$, and the price level is completely fixed in the short run. In what direction and by how much must government spending change to close the recessionary gap? Explain your thinking.
9. Suppose government spending increases. Would the effect on aggregate demand be larger if the Fed held the money supply constant in response or if the Fed committed to maintaining a fixed interest rate? Explain.
10. Is expansionary fiscal policy more likely to lead to a short-run increase in investment
- a. when the investment accelerator is large or when it is small? Explain.
- b. when the interest sensitivity of investment is large or when it is small? Explain.

11. Consider an economy described by the following equations:

$$Y = C + I + G$$

$$C = 100 + 0.75(Y - T)$$

$$I = 500 - 50r$$

$$G = 125$$

$$T = 100$$

where Y is GDP, C is consumption, I is investment, G is government purchases, T is taxes, and r is the interest rate. If the economy were at full employment (that is, at its natural level of output), GDP would be 2,000.

- a. Explain the meaning of each of these equations.
- b. What is the marginal propensity to consume in this economy?
- c. Suppose the central bank adjusts the money supply to maintain the interest rate at 4 percent, so $r = 4$. Solve for GDP. How does it compare to the full-employment level?
- d. Assuming no change in monetary policy, what change in government purchases would restore full employment?
- e. Assuming no change in fiscal policy, what change in the interest rate would restore full employment?

Chapter 34: The Influence of Monetary and Fiscal Policy on Aggregate Demand Problems and Applications

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